Light

GCSE Computer Science - Data Representation: Numbers (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What base is the hexadecimal number system?

- A. Base 2
- B. Base 8
- C. Base 10
- D. Base 16

Q2 (1 mark)

What are the only two digits used in binary?

- A. 0 and 1

- **B.** 1 and 2
- **C.** 0 and 9
- **D.** A and B

Q3 (1 mark)

What is a byte?

- **A.** A single binary digit
- **B.** A group of 4 bits
- **C.** A group of 8 bits
- **D.** A group of 16 bits

Q4 (1 mark)

What is the denary value of binary 1100?

- **A.** 12
- **B.** 10
- **C.** 6
- **D.** 14

Q5 (1 mark)

Which symbol correctly represents a byte?

- **A.** b
- **B.** B
- **C.** Bt
- **D.** by

Q6 (1 mark)

In hexadecimal, what denary value does the letter B represent?

- **A.** 10
- **B.** 13
- **C.** 12
- **D.** 11

Q7 (1 mark)

The same bit pattern can represent different types of data depending on context. What does this mean?

- **A.** Computers can only store one type of data
- **B.** The interpretation of bits depends on how the software uses them
- **C.** Binary numbers cannot represent text
- **D.** Each bit pattern has only one possible meaning

Q8 (1 mark)

What is the denary value of hexadecimal 20?

- ○ **A.** 20
- ○ **B.** 200
- ○ **C.** 40
- ○ **D.** 32

Q9 (1 mark)

What is the place value of the rightmost bit in any binary number?

- ○ **A.** 0
- ○ **B.** 8
- ○ **C.** 2
- ○ **D.** 1

Q10 (1 mark)

How many bytes are in 1 gigabyte (GB) using SI units?

- ○ **A.** 1 million bytes
- ○ **B.** 1 billion bytes
- ○ **C.** 100 million bytes

- ○ **D.** 10 million bytes

Q11 (1 mark)

What is the binary representation of denary 5?

- ○ **A.** 100
- ○ **B.** 111
- ○ **C.** 101
- ○ **D.** 110

Q12 (1 mark)

Which is larger: 500 bytes or 0.5 kB?

- ○ **A.** 500 bytes is larger
- ○ **B.** 0.5 kB is larger
- ○ **C.** They are equal
- ○ **D.** Cannot be compared

Q13 (2 marks)

Calculate: 1001 + 0110 in binary.

- ○ **A.** 1111

- B. 1110
- C. 1101
- D. 10000

Q14 (2 marks)

Convert hexadecimal 4B to binary.

- A. 0100 1010
- B. 0100 1011
- C. 0100 1100
- D. 0101 1011

Q15 (2 marks)

What is the result of left shifting 0000 1010 by 2 places?

- A. 0000 0010
- B. 0001 0100
- C. 0010 1000
- D. 0101 0000

Q16 (2 marks)

Convert denary 150 to binary.

- A. 1000 1110
- B. 1001 0100
- C. 1001 0110
- D. 1001 1000

Q17 (2 marks)

A photo is 4 MB. How many photos can fit on a 2 GB memory card?

- A. 500
- B. 200
- C. 50
- D. 2,000

Q18 (2 marks)

What is the result of right shifting 0110 0000 by 3 places?

- A. 0000 0110
- B. 1100 0000
- C. 0011 0000
- D. 0000 1100

Q19 (2 marks)

Convert denary 156 to hexadecimal.

- ○ **A.** 96
- ○ **B.** 9A
- ○ **C.** 9C
- ○ **D.** A6

Q20 (2 marks)

Calculate: 0100 + 0011 + 0010 in binary.

- ○ **A.** 1000
- ○ **B.** 1001
- ○ **C.** 1010
- ○ **D.** 1011

Q21 (2 marks)

Convert binary 1010 0111 to hexadecimal.

- ○ **A.** 97
- ○ **B.** A7
- ○ **C.** A9

- **D.** B7

Q22 (2 marks)

How many gigabytes are in 0.5 TB?

- **A.** 50 GB
- **B.** 100 GB
- **C.** 500 GB
- **D.** 5,000 GB

Q23 (2 marks)

To multiply a binary number by 16, how many left shifts are needed?

- **A.** 2
- **B.** 3
- **C.** 4
- **D.** 16

Q24 (3 marks)

Calculate the result of adding these three 8-bit binary numbers: 0010 0101 + 0001 1010 + 0010 0011

- **A.** 0101 1110

- **B.** 0110 0000
- **C.** 0110 0010
- **D.** 0110 0100

Q25 (3 marks)

A programmer sees an error code displayed as hexadecimal 7F. Convert this to binary and denary, then explain why hexadecimal is used for such codes.

- **A.** Binary: 0111 1111, Denary: 127. Hex is used because it provides a compact representation where each digit maps to exactly 4 bits
- **B.** Binary: 0111 1110, Denary: 126. Hex is used because computers process hexadecimal directly
- **C.** Binary: 1111 0111, Denary: 247. Hex is used because it requires fewer calculations
- **D.** Binary: 0111 1111, Denary: 128. Hex is used because it is the native computer language

Q26 (3 marks)

A game stores a character's health as binary 0100 0000 (64). The player takes damage that reduces health to one quarter. Explain how binary shifts can achieve this and state the result.

- **A.** Right shift by 2 gives 0001 0000 (16); each right shift divides by 2, so two shifts divide by 4, giving one quarter of 64
- **B.** Left shift by 2 gives 1 0000 0000 (256); shifts multiply health for bonus damage
- **C.** Right shift by 4 gives 0000 0100 (4); four shifts are needed for

quarter damage

- ○ **D.** Right shift by 1 gives 0010 0000 (32); one shift gives half, which is close to quarter

Q27 (3 marks)

An audio file is sampled at 44,100 samples per second with 16 bits per sample in stereo (2 channels). Calculate the file size for 1 minute of audio.

- ○ **A.** Approximately 5.3 MB
- ○ **B.** Approximately 10.6 MB
- ○ **C.** Approximately 21.2 MB
- ○ **D.** Approximately 42.4 MB

Q28 (3 marks)

Calculate: 0011 1001 + 0010 0111 + 0001 0100 in binary.

- ○ **A.** 0110 1110
- ○ **B.** 0110 1111
- ○ **C.** 0111 0000
- ○ **D.** 0111 0100

Q29 (3 marks)

Convert denary 189 to both binary and hexadecimal. Verify your answer by converting the hexadecimal back to denary.

- ○ **A.** Binary: 1011 1111, Hex: BF. Verify: B(11) x 16 + F(15) = 176 + 15 = 191
- ○ **B.** Binary: 1011 1011, Hex: BB. Verify: B(11) x 16 + B(11) = 176 + 11 = 187
- ○ **C.** Binary: 1011 1101, Hex: BD. Verify: B(11) x 16 + D(13) = 176 + 13 = 189
- ○ **D.** Binary: 1100 1101, Hex: CD. Verify: C(12) x 16 + D(13) = 192 + 13 = 205

Q30 (3 marks)

A student claims that 1,500,000 bytes is larger than 1.4 MB. Analyse this claim and determine if it is correct.

- ○ **A.** Correct: 1,500,000 bytes = 1.5 MB, which is larger than 1.4 MB
- ○ **B.** Incorrect: 1,500,000 bytes = 1.5 MB, which is smaller than 1.4 MB
- ○ **C.** Correct: 1,500,000 bytes = 15 MB, which is larger than 1.4 MB
- ○ **D.** Incorrect: 1,500,000 bytes = 0.15 MB, which is smaller than 1.4 MB

Total: **55** marks • Time: **30 min**

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